

KAILEY SIMONS

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

Candidate for M.Eng. in Computer Science, Data Science, Economics

May 2027

Candidate for B.S. in Computer Science, Data Science, and Economics, and B.S in Business Analytics

May 2026

Relevant Coursework: Statistical Learning Theory, Large-Scale Decision-Making & Inference, Design & Analysis of Algorithms, Econometrics Data Science, Statistics, Optimization in Business Analytics, Machine Learning

SKILLS & ACTIVITIES

Programming & Databases: Python, PostgreSQL, R, Julia, Java, TypeScript, HTML, STATA

Tools: AWS, Pandas, NumPy, Git, XGBoost, PyTorch, Scikit-learn, TensorFlow, Claude Code, Codex

Expertise: Data Science, Machine Learning (ML), Multi-Agent Systems, Statistical Modeling, AI Agents

Activities: MIT Varsity Swimming and Diving (All-American), HackMIT 2026

APPLIED ML RESEARCH EXPERIENCE

MIT Computer Science and Artificial Intelligence Laboratory

February 2026-Present

Applied ML Researcher

Cambridge, MA

- Analyzed ML models for Brazilian Portuguese audio deepfake detection; used RMS normalization to recover accuracy from 42% to 99.5% under codec compression (accepted to BRACIS; published in Springer Nature)
- Developed a framework for detecting and mitigating hallucinations in multi-agent financial QA systems, correcting up to 78.4% of detected hallucinations (published in MDPI)

MIT Center for Collective Intelligence

May 2024-June 2025

Undergraduate Researcher

Cambridge, MA

- Conducted A/B testing and statistical tests to evaluate human-AI interactions across 200+ participants

PROFESSIONAL EXPERIENCE

Accenture

June 2026-August 2026

Tech Architecture Analyst

New York City, NY

- Led agentic AI rollout strategy, architecting 3 AI agents projected to each save ~20 analyst hours per week
- Spearheaded development of 8+ Copilot adoption analytics, presenting insights directly to client executives

Balyasny Asset Management - MIT Generative AI Lab

February 2026-May 2026

AI Engineer

Cambridge, MA

- Built and tested MVPs of a multi-agent generation and evaluation pipeline to summarize Power & Gas markets, improving report quality by 59.4% and reducing the number of contradictions in reports by 73.3% on average

Nasdaq

January 2026-February 2026

Data Science & AI Engineer

Boston, MA

- Evaluated and implemented 2 literature-based models, reducing RMSE by 25%, to support Nasdaq's Dynamic M-ELO, the first AI-powered exchange order type
- Presented findings to the CTO and senior executives, contributing to the decision to advance the project

Vanguard

May 2025-August 2025

Investment Systems Software Engineer

Malvern, PA

- Spearheaded the integration of AI in the Portfolio Review Department's systems by designing an analytics system using AWS Bedrock, Lambda, and PostgreSQL, reducing analyst research time by 20%
- Partnered with engineers and analysts in an agile environment to deliver production-ready solutions, gaining end-to-end exposure to the software development lifecycle
- Developed a volatility nowcasting model using GARCH models and conducted exploratory data analysis

Daylit

January 2025-February 2025

AI Engineer

Boston, MA

- Engineered and deployed a loan review pipeline using the Google BERT model for Retrieval Augmented Generation (RAG) in Python, reducing manual review time by 50%+ and saving 300+ analyst hours per quarter

LEADERSHIP EXPERIENCE

MIT Undergraduate Economics Association

September 2024-May 2026

Co-President (2025-2026)/Publicity Chair (2024-2025)

Cambridge, MA

- Directed event strategy for 100+ students, increasing event participation by 10% through targeted publicity
- Led weekly meetings to plan events and drive improvements in the MIT undergraduate economics program

MIT Undergraduate Association Sustainability Committee

September 2023-May 2026

Co-President (2025-2026)/Publicity Lead (2023-2025)

Cambridge, MA

- Led weekly meetings to coordinate and host events with 5+ organizations across campus or from other schools